



**Sébastien
MARTINET**
10/07/1996
French

Summary

Data scientist with a PhD in physics and postdoctoral experience in large-scale data analysis, modeling, and uncertainty quantification. Skilled in building scalable data pipelines (Python, SQL, PySpark), applying machine learning (scikit-learn, PyTorch), and delivering reproducible workflows for complex datasets. Experienced in collaborative, international research projects and effective at translating technical insights into actionable results.

Languages: English ●●●● French (native) ●●●● Italian ●●○○ Spanish ●●○○

Numerical tools: Python, SQL, Fortran, C/C++, R, hTl5 & CSS

ML & Data Processing: Pandas, scikit-learn, Docker, Data Visualization (Matplotlib, seaborn), Dashboards (Streamlit, PowerBI), Pytorch, Tensorflow, PySpark, API, AWS

Soft Skills: Problem-solving, international talks, collaborative work, high-impact reports

✉ sebastien_martinet@ymail.com 🔗 sebastien.martinet.github.io

☎ Tel: +33 6 61 47 00 54 / 🏠 Address: Trieste, 34143, Italy

Work Experience

Oct 2022 – Present **PostDoctoral Researcher in Nuclear Astrophysics** [Université Libre de Bruxelles, BELGIUM](#)

Sept 2022 – Oct 2022 **PostDoctoral Researcher in Astrophysics** [University of Geneva, SWITZERLAND](#)

Sept 2018 – Aug 2022 **Doctoral Researcher in Astrophysics** [University of Geneva, SWITZERLAND](#)

Data Science Projects

Statistical Modeling & Scalable Data Pipelines [Python](#), [Fortran](#), [HPC](#), [Machine Learning](#) (7 publications - 3 first-author)

- Built a scalable pipeline to run and analyze large-scale simulations (10k+ runs) using legacy Fortran code wrapped in Python; automated deployment on HPC clusters.
- Developed and applied a statistical framework to quantify and propagate uncertainties, enabling robust predictions from high-dimensional data. Framework now used by international teams.
- Delivered reproducible workflows and end-to-end tooling to extract actionable insights from complex simulations.
- Collaborated with multidisciplinary teams to translate technical results into usable observables across domains.

Enhancing Legacy Software with Modern Data Science [Python](#), [Fortran](#), [Data Viz](#), [CLI Tools](#), [Git](#) (6 publications - 2 first-author)

- Led the design and integration of new physics modules into a legacy simulation code, improving performance and expanding capabilities for broader use cases.
- Modernized data analysis interface with Python and dashboards (NumPy, pandas, seaborn, Power BI, Streamlit), enabling rapid visual exploration.
- Developed robust CLI tools and notebooks for data wrangling and visualization, enhancing reproducibility and team collaboration.
- Demonstrated autonomy and leadership in driving complex projects from concept to publication.

Link to Dashboard Showcases [Python](#), [Streamlit](#), [API](#), [Plotly](#), [ML](#), [Time series analysis](#), [Energy and Finance](#)

Other projects (55 collaborators, 19 institutes, 11 countries) include: integrating models with large observational datasets, analyzing experimental data to improve simulations, publishing data products, and building scientific data platforms.

Education

2022

PhD in Astrophysics

University of Geneva, Switzerland

2018

Master in Astrophysics

Université Grenoble-Alpes, France

2016

Bachelor in Fundamental Physics

Université Pierre et Marie Curie, Paris VI, France

Selected Publications

[Martinet, S. et al. \(2025\)](#) Statistical framework for nuclear parameter uncertainties in nucleosynthesis modeling of r- and i-process. doi: [2025EPJA...61...48M](#)

[Martinet, S. et al. \(2024\)](#) The intermediate neutron capture process: IV. Impact of nuclear model and parameter uncertainties. doi: [2024A&6A...684A...8M](#)

[Full list of publications](#) / [Full list of conferences](#)

(22 papers incl. 9 first author) / (15 talks in 10 different countries)

Last updated: September, 2025.